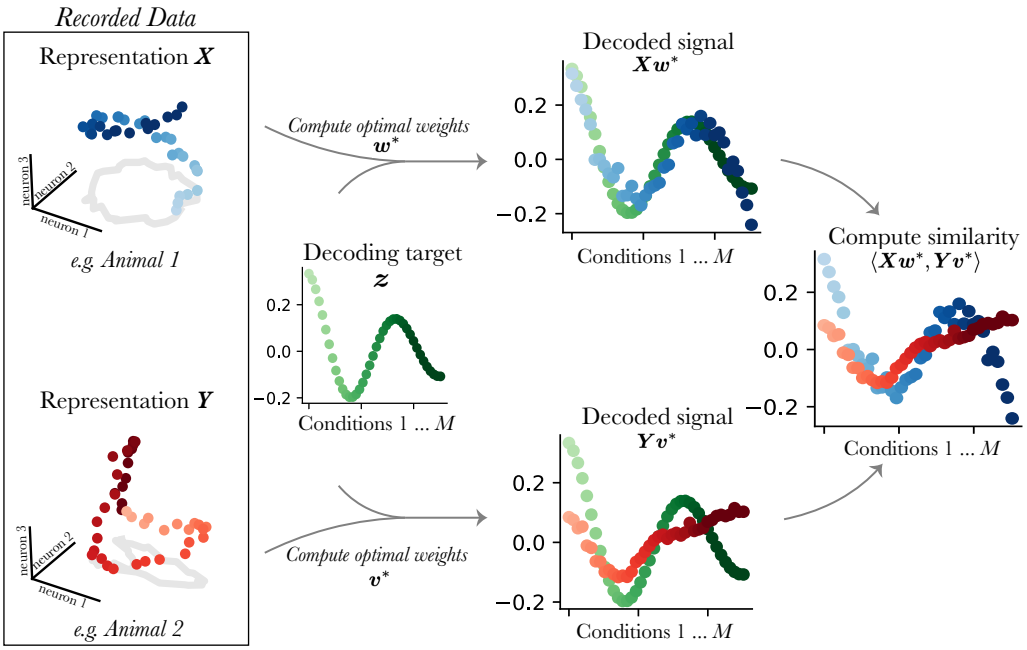

Figure 1: Schematic of the proposed framework for comparing representations $\boldsymbol{X}$ and $\boldsymbol{Y}$ (each dot represents mean neural responses to one of M conditions) in terms of a decoding target $\boldsymbol{z}$. First, optimal linear decoding weights $\boldsymbol{w}^*$ and $\boldsymbol{v}^*$ are computed. Then the similarity between the two systems is measured in terms of the *decoding similarity*: $\langle \boldsymbol{X}\boldsymbol{w}^*, \boldsymbol{Y}\boldsymbol{v}^* \rangle$.

Option 1 — best case alignment. An optimistic measure of representational similarity between networks is to find the decoding task $\boldsymbol{z} \in \mathbb{R}^M$ that results in a maximal decoder alignment. Formally,

$$\max_{\|\boldsymbol{z}\|_2=1} \langle \boldsymbol{X}\boldsymbol{w}^*, \boldsymbol{Y}\boldsymbol{v}^* \rangle = \max_{\|\boldsymbol{z}\|_2=1} \boldsymbol{z}^\top \boldsymbol{K}_X \boldsymbol{K}_Y \boldsymbol{z} \tag{9}$$

where the constraint that $\|\boldsymbol{z}\|_2 = 1$ is needed to keep the maximal value finite.

Option 2 — worst case alignment. An alternative is to find the task that minimizes alignment:

$$\min_{\|\boldsymbol{z}\|_2=1} \langle \boldsymbol{X}\boldsymbol{w}^*, \boldsymbol{Y}\boldsymbol{v}^* \rangle = \min_{\|\boldsymbol{z}\|_2=1} \boldsymbol{z}^\top \boldsymbol{K}_X \boldsymbol{K}_Y \boldsymbol{z} \tag{10}$$

which is obviously the pessimistic counterpart to option 1 above.

Option 3 — average case alignment. Instead of maximizing or minimizing over $\boldsymbol{z}$, we can approach the problem by averaging over a distribution of decoding tasks. Formally, we can quantify alignment in terms of the *average decoding similarity*:

$$\mathbb{E}\langle \boldsymbol{X}\boldsymbol{w}^*, \boldsymbol{Y}\boldsymbol{v}^* \rangle \tag{11}$$

where the expectation is taken with respect to $\boldsymbol{z} \sim P_{\boldsymbol{z}}$.

In the next subsection, we show that the third option is closely related to several existing neural representational similarity measures. However, all three quantities are potentially of interest and they are easy to compute as we document below in propositions 1 and 2.

Proposition 1. *The best case and worst case decoding alignment scores, eqs.* (9) *and* (10)*, are respectively given by the largest and smallest eigenvalues of:*

$$\frac{1}{2}\left(\boldsymbol{K}_X \boldsymbol{K}_Y + \boldsymbol{K}_Y \boldsymbol{K}_X\right) \tag{12}$$

Proposition 2. *The average decoding similarity, eq.* (11)*, is given by:*

$$\mathbb{E}\langle \boldsymbol{X}\boldsymbol{w}^*, \boldsymbol{Y}\boldsymbol{v}^* \rangle = \mathrm{Tr}\left(\boldsymbol{K}_X \boldsymbol{K}_z \boldsymbol{K}_Y\right), \tag{13}$$

where $\boldsymbol{K}_z := \mathbb{E}[\boldsymbol{z}\boldsymbol{z}^\top]$ *represents a kernel matrix for the targets.*

Similarity measure	a	b
Linear CKA	0	b
GULP	1	λ
CCA	1	0
ENSD	0	$\frac{1}{M}\frac{\mathrm{Tr}[\boldsymbol{C}_X^2]}{\mathrm{Tr}[\boldsymbol{C}_X]}$

Table 1: Existing similarity measures are equivalent to the decoding score in eq. (7) or the decoding distance in eq. (16) for different choices of a and b in eq. (5).

Proof of proposition 1. Let $\boldsymbol{A} \in \mathbb{R}^{M \times M}$ be any matrix, potentially non-symmetric. We can express $\boldsymbol{A}$ as a sum of a symmetric and skew symmetric matrix, $\boldsymbol{A} = \boldsymbol{B} + \boldsymbol{C}$, where:

$$\boldsymbol{B} = \frac{1}{2}\left(\boldsymbol{A} + \boldsymbol{A}^\top\right) \qquad \text{and} \qquad \boldsymbol{C} = \frac{1}{2}\left(\boldsymbol{A} - \boldsymbol{A}^\top\right) \tag{14}$$

For any $\boldsymbol{z} \in \mathbb{R}^M$, we have $\boldsymbol{z}^\top \boldsymbol{A}\boldsymbol{z} = \boldsymbol{z}^\top \boldsymbol{B}\boldsymbol{z} + \boldsymbol{z}^\top \boldsymbol{C}\boldsymbol{z}$. It is easy to verify that $\boldsymbol{z}^\top \boldsymbol{C}\boldsymbol{z}$ equals zero. Thus, $\boldsymbol{z}^\top \boldsymbol{A}\boldsymbol{z} = \boldsymbol{z}^\top \boldsymbol{B}\boldsymbol{z}$, and maximizing over $\boldsymbol{z}$ subject to $\|\boldsymbol{z}\|_2$ yields the largest eigenvalue of $\boldsymbol{B}$ (since it is symmetric). Likewise, minimizing yields the smallest eigenvalue of $\boldsymbol{B}$. Substituting $\boldsymbol{A} = \boldsymbol{K}_X \boldsymbol{K}_Y$ into this argument proves the proposition. □

Proof of proposition 2. Using eq. (7), we have:

$$\mathbb{E}\langle \boldsymbol{X}\boldsymbol{w}^*, \boldsymbol{Y}\boldsymbol{v}^* \rangle = \mathbb{E}[\boldsymbol{z}^\top \boldsymbol{K}_X \boldsymbol{K}_Y \boldsymbol{z}].$$

Using the cyclic property of the trace operator to manipulate this further, we get:

$$\boldsymbol{z}^\top \boldsymbol{K}_X \boldsymbol{K}_Y \boldsymbol{z} = \mathrm{Tr}(\boldsymbol{K}_X \boldsymbol{z}\boldsymbol{z}^\top \boldsymbol{K}_Y).$$

Taking the expectation with respect to $\boldsymbol{z}$ and using linearity of trace yields eq. (13). □

3 Interpretations of CKA and related measures

Intuitively, proposition 2 says that the expected inner product between optimal linear readouts is equal to a weighted matrix inner product between kernel matrices $\boldsymbol{K}_X$ and $\boldsymbol{K}_Y$ (with respect to a positive semi-definite weighting matrix $\boldsymbol{K}_z$, which of course depends on the distribution of decoding targets $\boldsymbol{z}$). Such kernel matrices and inner products are important for several methods of quantifying representational similarity. We can therefore leverage this result to provide new interpretations of several distance measures. The most basic idea is to consider a distribution over $\boldsymbol{z}$ which satisfies $\boldsymbol{K}_z = \boldsymbol{I}$. This leads us to measure distances using standard Euclidean geometry, which we state as the following corollary to proposition 2. We explore a relaxation of this assumption and the limit as the number of input samples $M \to \infty$ in appendix A.3.

Corollary 1. *For any distribution over decoding targets satisfying $\boldsymbol{K}_z = \boldsymbol{I}$, the Frobenius inner product between normalized kernel matrices $\boldsymbol{K}_X$ and $\boldsymbol{K}_Y$ is equal to the average decoding similarity:*

$$\mathbb{E}\langle \boldsymbol{X}\boldsymbol{w}^*, \boldsymbol{Y}\boldsymbol{v}^* \rangle = \mathrm{Tr}[\boldsymbol{K}_X \boldsymbol{K}_Y] \tag{15}$$

Further, when $\boldsymbol{K}_z = \boldsymbol{I}$, the (squared) average decoding distance is equal to the (squared) Euclidean distance between $\boldsymbol{K}_X$ and $\boldsymbol{K}_Y$:

$$\mathbb{E}\|\boldsymbol{X}\boldsymbol{w}^* - \boldsymbol{Y}\boldsymbol{v}^*\|_2^2 = \|\boldsymbol{K}_X - \boldsymbol{K}_Y\|_F^2. \tag{16}$$

This corollary can be used to unify several neural representational similarity measures, each corresponding to a different choice of the penalty function $\boldsymbol{G}(\boldsymbol{X})$, Table 1. Further, it yields an interpretation of each measure in terms of the expected overlap or difference in decoding readouts between networks.

Linear CKA First, for $b > 0$, the following choice of $\boldsymbol{G}(\boldsymbol{X})$ yields the linear CKA score [6, 21]:

$$\boldsymbol{G}(\boldsymbol{X}) = b\boldsymbol{I} \quad \Rightarrow \quad \frac{\mathbb{E}\langle \boldsymbol{X}\boldsymbol{w}^*, \boldsymbol{Y}\boldsymbol{v}^* \rangle}{\sqrt{\mathbb{E}\langle \boldsymbol{X}\boldsymbol{w}^*, \boldsymbol{X}\boldsymbol{w}^* \rangle \mathbb{E}\langle \boldsymbol{Y}\boldsymbol{v}^*, \boldsymbol{Y}\boldsymbol{v}^* \rangle}} = \frac{\mathrm{Tr}[\boldsymbol{K}_X \boldsymbol{K}_Y]}{\|\boldsymbol{K}_X\|_F \|\boldsymbol{K}_Y\|_F} = \mathrm{CKA}(\boldsymbol{X}, \boldsymbol{Y})$$

where we have normalized the expected inner product to obtain a similarity score ranging between 0 and 1. Note that the centering step of CKA is taken care of by our assumption, stated at the beginning of section 2, that the columns of $\boldsymbol{X}$ and $\boldsymbol{Y}$ are preprocessed to have mean zero. Thus, linear CKA be interpreted as a normalized average decoding similarity with quadratic weight regularization.

GULP distance The CKA score is closely related to Euclidean distance, so it is not surprising that we can extend our analysis to other methods that utilize this distance. For example, the sample GULP distance, proposed by [3], is the equal to the average decoding distance with a regularization parameter $\lambda > 0$. In particular,

$$\boldsymbol{G}(\boldsymbol{X}) = \boldsymbol{C}_X + \lambda \boldsymbol{I} \quad \Rightarrow \quad \mathbb{E}\|\boldsymbol{X}\boldsymbol{w}^* - \boldsymbol{Y}\boldsymbol{v}^*\|_2^2 = \|\boldsymbol{K}_X - \boldsymbol{K}_Y\|_F^2 = \mathrm{GULP}_\lambda^2(\boldsymbol{X}, \boldsymbol{Y})$$

yields the plug-in estimate of the squared GULP distance [3, section 3]. We therefore see that the Euclidean distance between normalized kernel matrices can be interpreted as equal to the average decoding distance, or also as an approximation of the population formulation of the GULP distance established in [3].

Canonical correlation analysis (CCA) CCA is another method that has been used to compare neural representations that fits nicely into our framework [28, 35]. When the two networks have the same dimension, i.e. $N = N_X = N_Y$, the output of CCA is a sequence of N canonical correlation coefficients $1 \geq \rho_1 \geq \cdots \geq \rho_N \geq 0$. The average squared canonical coefficients is sometimes used as a measure of similarity,[1] and Kornblith et al. [21] showed this to be related to the linear CKA score on whitened representations. In particular, assuming the covariance matrices $\boldsymbol{C}_X$ and $\boldsymbol{C}_Y$ are invertible, we have:

$$\boldsymbol{G}(\boldsymbol{X}) = \boldsymbol{C}_X \quad \Rightarrow \quad \frac{\mathbb{E}\langle \boldsymbol{X}\boldsymbol{w}^*, \boldsymbol{Y}\boldsymbol{v}^* \rangle}{\sqrt{\mathbb{E}\langle \boldsymbol{X}\boldsymbol{w}^*, \boldsymbol{X}\boldsymbol{w}^* \rangle \mathbb{E}\langle \boldsymbol{Y}\boldsymbol{v}^*, \boldsymbol{Y}\boldsymbol{v}^* \rangle}} = \frac{1}{N} \sum_{i=1}^{N} \rho_i^2 \tag{17}$$

In practice, it is often useful to incorporate regularization into CCA because the covariance matrices $\boldsymbol{C}_X$ and $\boldsymbol{C}_Y$ may be ill-conditioned. As mentioned above, the regularization used in GULP distance corresponds to choosing $\boldsymbol{G}(\boldsymbol{X}) = \boldsymbol{C}_X + \lambda \boldsymbol{I}$. A similar approach proposed in [35] is to use $\boldsymbol{G}(\boldsymbol{X}) = (1-\alpha)\boldsymbol{C}_X + \alpha \boldsymbol{I}$ for a hyperparameter $0 \leq \alpha \leq 1$. This effectively allows for a continuous interpolation between a CCA-based similarity score and linear CKA.

Effective Number of Shared Dimensions (ENSD) Finally, we note a connection to recent work by Giaffar et al. [15], who studied the following quantity, which they call the ENSD:

$$\mathrm{ENSD}(\boldsymbol{X}, \boldsymbol{Y}) = \frac{\mathrm{Tr}[\boldsymbol{X}\boldsymbol{X}^\top]\,\mathrm{Tr}[\boldsymbol{Y}\boldsymbol{Y}^\top]\,\mathrm{Tr}[\boldsymbol{X}\boldsymbol{X}^\top\boldsymbol{Y}\boldsymbol{Y}^\top]}{\mathrm{Tr}[\boldsymbol{X}\boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{X}^\top]\,\mathrm{Tr}[\boldsymbol{Y}\boldsymbol{Y}^\top\boldsymbol{Y}\boldsymbol{Y}^\top]} \tag{18}$$

This is simply a rescaling of the inner product, $\mathrm{Tr}[\boldsymbol{X}\boldsymbol{X}^\top\boldsymbol{Y}\boldsymbol{Y}^\top]$ and therefore easily fits within our framework using the following choice for $\boldsymbol{G}(\boldsymbol{X})$:

$$\boldsymbol{G}(\boldsymbol{X}) = \frac{\mathrm{Tr}[\boldsymbol{C}_X^2]}{\mathrm{Tr}[\boldsymbol{C}_X]}\boldsymbol{I} \quad \Rightarrow \quad \mathbb{E}\langle \boldsymbol{X}\boldsymbol{w}^*, \boldsymbol{Y}\boldsymbol{v}^* \rangle = \mathrm{ENSD}(\boldsymbol{X}, \boldsymbol{Y}) \tag{19}$$

One of the most interesting features of ENSD is it's connection to the **participation ratio**, $\mathcal{R}$:

$$\mathcal{R}(\boldsymbol{C}_X) = \frac{(\mathrm{Tr}[\boldsymbol{C}_X])^2}{\mathrm{Tr}[\boldsymbol{C}_X^2]} = \mathrm{ENSD}(\boldsymbol{X}, \boldsymbol{X}) \tag{20}$$

which is used within physics and computational neuroscience (e.g. [14, 7, 30]) as a continuous analogue to the rank of a matrix, and is a measure of the effective dimensionality of the data matrix

[1] In the statistics literature, this summary statistic is sometimes called *Yanai's generalized generalized coefficient of determination* [29, 37].